

Determinants of cyclic covers of the line: a finite-quotient reduction to Fermat varieties

Kyle Kistner

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Abstract

Let $C \rightarrow \mathbf{P}^1$ be a cyclic cover of prime degree p with $s \geq 3$ branch points. Put $h = s - 2$. We exhibit the rational cyclotomic determinant of $H^1(C)$ as an algebraic direct summand of the middle cohomology of a degree- p Fermat hypersurface of dimension h . The construction uses two finite maps to the same normal quotient and an explicit Chow-correspondence identity; it requires no specialization or singular-cohomology comparison. Shioda's published theorem for arbitrary products of prime-degree Fermat varieties then implies algebraicity of every rational Hodge class in every tensor product of these determinant spaces. In particular it implies the balanced mixed-determinant statement for arbitrary lists of cyclic covers. This supplies an attribution-sensitive simplification of the companion fusion argument. The Fermat-quotient construction has direct precedent in Schoen's 1989 paper; no priority claim is made for the construction or its product consequence. This is a classical proof draft, not a formalization or an independent referee certification.

1 Statement and exact rational realization

All varieties and correspondences are over $\mathbf{C}$, and Chow groups have rational coefficients. Fix a prime p and put $K = \mathbf{Q}(\zeta_p)$. The argument includes $p = 2$, although the principal application has p odd. Let $C \rightarrow \mathbf{P}^1$ be a smooth connected cyclic degree- p cover with chosen deck generator σ and distinct branch points $t_1, \dots, t_s$. Write its nonzero residues as $a_i \in \mathbf{F}_p^\times$, with $\sum_i a_i = 0$. Assume $s \geq 3$, and set $h = s - 2$ and $J = J(C)$. Then $\dim_K H^1(C, \mathbf{Q}) = h$ and the genus is $(p - 1)h/2$. The action of K sends ζ_p to σ^* ; use the compatible action on J under the Abel–Jacobi identification of H^1 .

The rational determinant realization D_C is specified by

$$(D_C)_{\mathbf{C}} = \bigoplus_{b=1}^{p-1} \bigwedge^h H^1(C, \mathbf{C})_b, \quad \dim_{\mathbf{Q}} D_C = p - 1.$$

It has two natural realizations: in $H^h(C^h, \mathbf{Q})$ by antisymmetrized Künneth tensors, and in

$$H^h(J, \mathbf{Q}) = \bigwedge^h H^1(J, \mathbf{Q}).$$

An algebraic correspondence between these realizations is explained below. Write

$$F_p^h = \{x_1^p + \dots + x_{h+2}^p = 0\} \subset \mathbf{P}^{h+1}.$$

Theorem 1.1. *There is a rational algebraic projector on F_p^h whose cohomological image, concentrated in degree h , is isomorphic to D_C by algebraic correspondences in both directions. For the realization in C^h , the projector and the two inverse correspondences already satisfy the required identities in rational Chow groups. In particular the realization of D_C in $H^h(J, \mathbf{Q})$ is the image of a rational algebraic correspondence from $H^h(F_p^h, \mathbf{Q})$.*

Corollary 1.2. *Let $C_1, \dots, C_N$ be any finite list of such covers of the same prime degree, with arbitrary branch points, residues and repetitions. Put $h_j = s_j - 2$ and $H = \sum_j h_j$. Every rational Hodge class in*

$$\bigotimes_{\mathbf{Q}, j} D_{C_j} \subset H^H \left(\prod_j J(C_j), \mathbf{Q} \right)$$

is algebraic. The same holds in any rational algebraically projected subspace of this tensor product, including the subspace with all cyclotomic embedding labels equal.

No monodromy, genericity or endomorphism-algebra assumption appears here. The assertion is about determinant subspaces, not the entire cohomology of an arbitrary Jacobian or its powers.

2 A common normal finite quotient

Choose a coordinate on $\mathbf{P}^1$ for which all t_i are finite. With representatives $1 \leq a_i \leq p-1$, the cover has a Kummer equation

$$y^p = \prod_{i=1}^{h+2} (x - t_i)^{a_i}.$$

The point at infinity is unramified because $\sum_i a_i = pk$ for an integer k . This equation describes the function field; C denotes its smooth projective normalization.

Let $Z = C^h$. On its h factors act μ_p^h , and let

$$N = \{(\epsilon_1, \dots, \epsilon_h) \in \mu_p^h : \epsilon_1 \cdots \epsilon_h = 1\}, \quad G = N \rtimes \mathfrak{S}_h.$$

Permutations permute the factors and normalize N . Let $Y = Z/G$, the normal projective finite quotient. Its degree over $\text{Sym}^h \mathbf{P}^1 = \mathbf{P}^h$ is p .

On the dense affine locus write $u(T) = \prod_{\nu=1}^h (T - x_\nu)$. Its coefficients are coordinates on $\text{Sym}^h \mathbf{A}^1$. The N -invariant element $z = \prod_{\nu=1}^h y_\nu$ is permutation invariant, and, after rescaling z by a constant, satisfies

$$z^p = \prod_{i=1}^{h+2} u(t_i)^{a_i}. \tag{1}$$

The coordinate-field extension of the ordered product has group μ_p^h over $\mathbf{C}(x_1, \dots, x_h)$: the h Kummer classes are independent, as seen by valuation at $x_\nu = t_i$. Thus taking N -invariants leaves exactly the field generated by z . Taking permutation invariants gives exactly (1) over the symmetric coefficient field. This also follows by comparing the degree p of that equation with the index of G in $\mu_p^h \rtimes \mathfrak{S}_h$.

Homogeneously, evaluation at t_i is a linear form L_i on $H^0(\mathbf{P}^1, \mathcal{O}(h))$. The $h+2$ hyperplanes $L_i = 0$ are in general position: any $h+1$ evaluations at distinct points are independent by polynomial interpolation. Consequently there is a unique linear relation, up to scaling,

$$\sum_{i=1}^{h+2} \lambda_i L_i = 0, \quad \lambda_i \neq 0 \quad \text{for every } i.$$

The nonvanishing follows because a relation omitting one evaluation would contradict that independence.

Consider the smooth diagonal hypersurface

$$X = \{\sum_i \lambda_i x_i^p = 0\} \subset \mathbf{P}^{h+1}.$$

Over $\mathbf{C}$ it is isomorphic to F_p^h by diagonal rescaling. The map $[x_i] \mapsto [x_i^p]$ is a finite quotient onto the hyperplane $\sum_i \lambda_i u_i = 0$, identified with $\mathbf{P}^h$ by $u_i = L_i$. Its group is

$$A = \mu_p^{h+2} / \mu_p^{\text{diag}}.$$

Because $\sum_i a_i = 0$ modulo p , the formula $\chi_a((\epsilon_i)) = \prod_i \epsilon_i^{a_i}$ is a well-defined nontrivial character of A . Put $B = \ker \chi_a$; then $|B| = p^h$. The quotient X/B has precisely the Kummer function field (1). For example, in the chart $x_{h+2} \neq 0$, the semi-invariant function

$$w = \frac{\prod_i x_i^{a_i}}{x_{h+2}^{pk}} \quad \text{satisfies} \quad w^p = \frac{\prod_i u_i^{a_i}}{u_{h+2}^{pk}}.$$

Since both quotients are normal finite covers of the same $\mathbf{P}^h$ with the same function field, they are isomorphic to its normalization in that field. We have therefore constructed finite quotient maps

$$r : X \longrightarrow Y \longleftarrow Z : q, \quad d_X = \deg r = p^h, \quad d_Z = \deg q = p^{h-1}h!. \quad (2)$$

The intermediate Y may be singular. This causes no difficulty for the next argument, which uses only cycles on smooth X and Z .

3 An elementary finite-correspondence identity

For smooth projective varieties of equal dimension h , a correspondence in $\text{CH}^h(X \times Z)$ acts from the cohomology of X to that of Z ; write T^t for its transpose. Let $[X \times_Y Z]$ mean the dimension- h fundamental cycle of the scheme-theoretic fiber product, with its generic multiplicities.

Lemma 3.1. *In the situation (2), set $T = [X \times_Y Z]$. Then*

$$\begin{aligned} T \circ T^\text{t} &= d_X \sum_{g \in G} [\Gamma_g] && \text{in } \text{CH}^h(Z \times Z), \\ T^\text{t} \circ T &= d_Z \sum_{b \in B} [\Gamma_b] && \text{in } \text{CH}^h(X \times X). \end{aligned}$$

Proof. The intersection support in each composition is finite over Y . In particular the two pulled-back correspondences intersect properly in $Z \times X \times Z$ (or $X \times Z \times X$): the intersection has dimension h . On a dense open subset of Y both quotient maps are finite étale torsors. There the composition from Z to itself consists of each graph of $g \in G$ with multiplicity d_X : for a point and its g -translate one can choose any of the d_X points of X above their common image. The other identity is identical with X and Z interchanged.

These generic computations determine the entire dimension- h cycles. Indeed, a component supported above the excluded proper closed subset of Y has dimension at most $h - 1$, since all the relevant maps are finite. It cannot contribute a dimension- h component to the proper intersection or its pushforward. Thus the generic cycle identities hold globally, and therefore also in the indicated Chow groups. No pushforward to singular cohomology or resolution theorem has been used. $\square$

4 The determinant projector and transfer

For a finite group L acting on Z , write $e_L = |L|^{-1} \sum_{\ell \in L} [\Gamma_\ell]$. Let $\tilde{G} = \mu_p^h \rtimes \mathfrak{S}_h$, containing G as a normal subgroup of index p , and set

$$P = e_G - e_{\tilde{G}} \in \mathrm{CH}^h(Z \times Z). \quad (3)$$

Nested averaging gives $e_G e_{\tilde{G}} = e_{\tilde{G}} e_G = e_{\tilde{G}}$, hence $P^2 = P$. This formula uses the quotient deck action through one slot; it never replaces it by simultaneous action on all slots. In particular it remains valid when $p \mid h$.

Lemma 4.1. *The only nonzero cohomological image of P is in degree h , where*

$$\mathrm{im} P = \bigoplus_{b=1}^{p-1} \bigwedge^h H^1(C, \mathbf{C})_b$$

after extension of scalars to $\mathbf{C}$.

Proof. Künneth tensors invariant under N have equal deck-character labels on their h slots, because the characters trivial on N are exactly $(b, \dots, b)$. The subtraction of $e_{\tilde{G}}$ removes the label $b = 0$. A nontrivial character occurs only in $H^1(C)$, because the deck action on $H^0(C)$ and $H^2(C)$ is trivial. Thus every surviving tensor has total degree h and lies in $H^1(C)_b^{\otimes h}$.

A geometric transposition of two factors acts on such Künneth tensors as minus the ordinary tensor transposition, by graded commutativity. Consequently taking geometric $\mathfrak{S}_h$ -invariants takes ordinary alternating tensors, namely $\bigwedge^h H^1(C)_b$. Each is a line since $\dim H^1(C)_b = h$. The rationality is automatic from (3). $\square$

Put $d = d_X d_Z$ and define

$$R = \frac{1}{d} T^t P T \in \mathrm{CH}^h(X \times X), \quad a = P T, \quad b = \frac{1}{d} T^t P.$$

Lemma 3.1 and $P e_G = P$ give

$$R^2 = R, \quad ab = P, \quad ba = R.$$

For instance $PTT^t P = dP$. These are Chow identities. They exhibit (Z, P) as a direct summand of X with inverse algebraic correspondences. Lemma 4.1 proves the first realization of Theorem 1.1.

Passage to the Jacobian. Let $\iota : C \rightarrow J$ be an Abel–Jacobi map based at a branch point; it is equivariant for the stated deck convention and induces an isomorphism $\iota^* : H^1(J, \mathbf{Q}) \rightarrow H^1(C, \mathbf{Q})$. Its inverse is algebraic in degree one. Here is a direct way to make this familiar fact sufficient for our use. The inverse Hodge map, using the intersection pairing on $H^1(C)$, is a rational Hodge class in the Künneth component $H^1(C, \mathbf{Q}) \otimes H^1(J, \mathbf{Q}) \subset H^2(C \times J, \mathbf{Q})$. By the Lefschetz (1, 1) theorem it is the class of a rational divisor. Composing this divisor correspondence with the curve projector

$$\pi_1^C = [\Delta_C] - [c_0 \times C] - [C \times c_0]$$

ensures its relevant source is precisely $H^1(C)$. Call the resulting correspondence $L : C \rightsquigarrow J$.

Take $L^{\otimes h}$ from C^h to J^h and follow it by pullback along the diagonal $J \rightarrow J^h$. On the alternating subspace this is the usual wedge map. With the alternating inclusion normalized by $1/h!$, it identifies the determinant realization in C^h with that in J . All maps are algebraic. For an explicit inverse on the determinant, pull back by the sum map $C^h \rightarrow J$, $(c_1, \dots, c_h) \mapsto \sum_\nu \iota(c_\nu)$, apply $(\pi_1^C)^{\otimes h}$, and divide by $h!$. Its degree- h component is the normalized alternating inclusion followed by $(\iota^*)^{\otimes h}$. This proves the remaining assertion of Theorem 1.1. No Hodge conjecture in higher codimension is used in this degree-one passage.

5 Products, rational Hodge classes, and balance

We use the following exact published input.

Theorem 5.1 (Shioda). *For a fixed prime p , the rational Hodge conjecture holds for every finite product $\prod_j F_p^{h_j}$, with arbitrary dimensions h_j .*

This is Theorem 2 on printed page 112 of [1]; the paragraph immediately preceding it verifies the condition used there for prime degree and every dimension. The degree is fixed, but the dimensions need not agree. This is stronger than the single-Fermat-hypersurface assertion.

To prove Corollary 1.2, first use the inverse degree-one correspondences to place a rational Hodge class in $\bigotimes_j \text{im } P_j \subset H^H(\prod_j C_j^{h_j}, \mathbf{Q})$. The exterior product of the b_j constructed above maps it to a rational Hodge class on $\prod_j F_p^{h_j}$; the exterior product of the a_j returns the original class. If H is even, Shioda makes the source class algebraic, and its image under the forward algebraic correspondence is algebraic. If H is odd there is no rational (r, r) class in degree H . This proves the corollary. The explicit algebraic splitting means that even semisimplicity of polarizable Hodge structures is unnecessary for this proof.

For comparison with the companion fusion statement, the matched space is

$$W_{\mathbf{C}} = \bigoplus_{b=1}^{p-1} \bigotimes_j \bigwedge^{h_j} H^1(C_j, \mathbf{C})_b.$$

It is rational. To realize matching by an algebraic projector, one may work in all $H = \sum h_j$ curve slots and use

$$\prod_{r=2}^H \left(\frac{1}{p} \sum_{u=0}^{p-1} (\sigma_1^*)^u (\sigma_r^*)^{-u} \right).$$

On the already alternating factors it selects exactly equal labels in all slots. This formulation also works when one or several h_j are divisible by p ; simultaneous action on h_j slots is not substituted for the action on one selected slot.

Write $[a]_p \in \{1, \dots, p-1\}$ for the representative of a nonzero residue, and put $S = \sum_j s_j$. The eigenspace Hodge numbers of a cyclic cover give the criterion

$$2 \sum_{j,i} [ba_{ji}]_p = pS \quad (b \in \mathbf{F}_p^\times) \quad (4)$$

for every component line of W to have bidegree $(H/2, H/2)$. Changing the pullback labeling interchanges b with $-b$, leaving the criterion unchanged. Hence (4) and Corollary 1.2 imply

$$W \subset \text{im} \left(\text{cl} : \text{CH}^{H/2} \left(\prod_j J(C_j) \right)_{\mathbf{Q}} \longrightarrow H^H \left(\prod_j J(C_j), \mathbf{Q} \right) \right).$$

There is no need to decompose a balanced residue tuple into opposite pairs, construct a degeneration, or assume that an individual determinant is a Hodge class. Jointly balanced products of non-Hodge determinants are included directly.

Remark 5.2 (A previously proposed test example). For $p = 5$, the branch tuples $(1, 1, 3)$ and $(1, 2, 4, 4, 4)$ give $h_1 = 1$ and $h_2 = 3$. Their residue-age vectors, for multipliers $1, 2, 3, 4$, are $(1, 1, 2, 2)$ and $(3, 3, 2, 2)$. Their sums are all four, so the matched rational four-dimensional space in $H^1(J(C_1)) \otimes H^3(J(C_2))$ has type $(2, 2)$. Its algebraicity is a consequence of the transfer from $F_5^1 \times F_5^3$ and Shioda's theorem. The fact that neither individual determinant is of Tate type does not distinguish this example from the Fermat-product method.

6 Attribution and limits

The cycle theorem doing the essential algebraicity work is Shioda’s published Fermat-product theorem [1]. Schoen’s 1989 paper [2], in its directly inspected introduction on printed pages 137–138, announces algebraic correspondences from cyclic covers of $\mathbf{P}^v$ branched over $v + 2$ hyperplanes, which are Fermat quotients, to determinant Hodge structures of cyclic-cover generalized Pryms. That is direct precedent for the construction used here, and must be acknowledged prominently.

At the time of writing, only those two introduction pages of [2] were available to this audit; the exact hypotheses of its later sections were not inspected. Accordingly we do not cite an unverified numbered theorem as if it already had precisely the scope of Theorem 1.1. Instead Sections 2–4 give the needed $\mathbf{P}^1$ argument explicitly. This access limitation does not become a claim of novelty. The product consequence is a short deduction from classical Fermat theory once this finite-quotient correspondence is written down; identifying its earliest explicit publication remains a bibliographic task.

The companion all-powers assertion for a very general elementary-abelian cover additionally needs the simultaneous monodromy classification and the algebraic invariant-generator reduction. Those assertions do not follow merely from this note. Conversely the arbitrary balanced determinant algebraicity step should not be described as a new cycle-construction breakthrough on the evidence presently available.

References

- [1] T. Shioda, *The Hodge conjecture and the Tate conjecture for Fermat varieties*, Proc. Japan Acad. Ser. A Math. Sci. **55** (1979), 111–114, Theorem 2, p. 112. doi:10.3792/pjaa.55.111. Author-uploaded original scan (direct PDF). The landing-page metadata labels the later Math. Ann. paper; the uploaded four-page primary scan is the Proc. Japan Acad. note cited here.
- [2] C. Schoen, *Cyclic covers of $\mathbf{P}^v$ branched along $v + 2$ hyperplanes and the generalized Hodge conjecture for certain abelian varieties*, in Arithmetic of Complex Manifolds, Lecture Notes in Math. **1399**, Springer, 1989, 137–154. doi:10.1007/BFb0095974. Publisher introduction preview.
- [3] C. Schoen, *Hodge classes on self-products of a variety with an automorphism*, Compositio Math. **65** (1988), 3–32. Original journal scan.
- [4] N. Aoki, *Hodge cycles on CM abelian varieties of Fermat type*, Comment. Math. Univ. Sancti Pauli **51** (2002), 99–130, Theorem 1.2, p. 102. Original university-hosted PDF.